

RFM Analysis Report: Customer Purchase Behavior Segmentation

Project: RFM-Based Customer Analytics (2025 Dataset)

Analysis Date: 21 March 2026

1. Executive Summary

This project applied RFM (Recency, Frequency, Monetary) analysis to the complete 2025 customer transaction dataset, consisting of 12 monthly tables. The primary objective was to quantify customer engagement by evaluating how recently customers purchased, how often they buy, and their total expenditure. This data was then used to automatically segment the base into 10 actionable groups ranging from "Can't Lose Them" (top-tier customers) to "Lost" (inactive).

Key business metrics at a glance:

- Total Customers: 287
- Total Orders: 994
- Total Sales: ₹17.07K
- Average Order Value (AOV): ₹3.46

The segmentation reveals a classic long-tail distribution: a small group of highly valuable customers and a large base of lower-engagement customers. This provides a clear roadmap for retention, re-activation, and strategic growth campaigns.

2. Methodology – How RFM Scores Were Built

The entire pipeline was implemented in SQL (BigQuery) for maximum scalability and reproducibility. The process followed a structured sequential flow:

1. **Data Union:** All 12 monthly tables (2025_01 to 2025_12) were combined into a single unified table named `rfm_2025`.

- 2. **RFM Calculation:** Metrics were derived using the following logic:
 - o **Recency:** Days elapsed since the last purchase relative to the analysis date.
 - o **Frequency:** Total count of orders per unique customer.
 - o **Monetary:** Rounded sum of total order value per customer.
- 3. **Ranking:** Individual row numbers were assigned for each metric (Recency in ascending order; Frequency and Monetary in descending order).
- 4. **Scoring:** A 1–10 scale was applied using NTILE functions. Each customer received a score for each category, where higher numbers represent more desirable behavior.
- 5. **RFM Total:** The three scores were aggregated for a maximum possible total of 30.

Segment Labeling Logic:

Customers were categorized into segments based on their `rfm_total` score, ensuring fully automated and objective classification without manual intervention.

3. Customer Segment Distribution

The following distribution highlights the headcount and behavioral characteristics of each segment based on the 2025 dataset:

Rank	Segment	Customers	% of Base	Typical Behavior
1	Promising	50	17.4%	Moderate recent activity, low spend
2	Hibernating	38	13.2%	Inactive for long time
3	Potential Loyalist	34	11.8%	Good frequency & recency, building value
4	Needs Attention	33	11.5%	Declining activity
5	Repeat Buyers	30	10.5%	Consistent but lower spend

Rank	Segment	Customers	% of Base	Typical Behavior
6	Champions	29	10.1%	High scores across all three metrics
7	Loyal VIP	29	10.1%	Very high spend & frequency
8	At Risk	25	8.7%	Recent drop in activity
9	Can't Lose Them	12	4.2%	Perfect 29–30 RFM score – highest value
10	Lost	7	2.4%	No activity for very long time

Insight: The top three highest-value segments ("Can't Lose Them," "Champions," and "Loyal VIP") represent only 24.4% of the customer base (70 individuals) but are responsible for the vast majority of total revenue.

4. Deep Dive into Top Customers

The analysis identifies specific high-scoring customers (RFM Total 28–30) who anchor the business revenue.

- **CUST0012:** Recency 80 days, Frequency 9, Monetary ₹187.09 (RFM Total: 30)
- **CUST0215:** Recency 84 days, Frequency 6, Monetary ₹123.08 (RFM Total: 30)

These customers exhibit the "rock star" profile: they have purchased recently, they buy often, and they spend significantly more than the average. Protecting these relationships is the organization's highest priority.

5. Strategic Recommendations by Segment

Segment	Action Priority	Recommended Tactics
Can't Lose Them	★★★★★ ▾	VIP-only events, personalized offers, early access, dedicated account manager
Champions / Loyal VIP	★★★★☆ ▾	Loyalty tier upgrades, referral bonuses, cross-sell high-margin products
Potential Loyalist / Repeat Buyers	★★★★☆ ▾	Welcome-back campaigns, bundle offers, win-back discounts
Promising / Needs Attention	★★★★☆ ▾	Re-engagement emails, limited-time 10–15% off, educational content
At Risk / Hibernating	★★★☆☆ ▾	"We miss you" campaigns, survey for churn reasons, aggressive win-back
Lost	★★☆☆☆ ▾	One final reactivation email (low cost) – otherwise de-prioritize

6. Business Impact & Next Steps

- **Immediate Value:** Marketing and sales teams now possess a single source of truth for customer segmentation that can be refreshed automatically.
- **Expected ROI:** By focusing 10–15% of the marketing budget on top-tier segments and successfully re-activating 20% of "At Risk" customers, the business can expect a significant revenue lift with minimal additional expenditure.
- **Future Enhancements:**

- a. Incorporate Average Order Value (AOV) and Customer Tenure as additional scoring dimensions.
- b. Develop an automated dashboard in Google Looker Studio with interactive drill-down capabilities.
- c. Schedule monthly refresh jobs and configure email alerts for significant segment shifts.

7. Conclusion

The RFM analysis successfully transformed raw transaction data into a strategic asset. Even with a modest base of 287 customers, the scoring reveals massive variances in value and behavior. By adopting these insights, the business can transition from broad "spray-and-pray" marketing to precision targeting—retaining high-value customers and efficiently reviving dormant ones.

Project Status: Completed (SQL pipeline and dashboard ready for deployment).